TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Metric	Consolidated Value
Avg Fidelity	0

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Dataset Audit: breast_cancer_wisconsin_diagnostic.csv

1. Dataset Portrait

Attribute	Value
Rows	569
Columns	31
Numeric features	30
Categorical features	1
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4.1-nano
Folds (*13)	0
Context % (*14)	20%
Generation % (*15)	80%
Join context (*16)	True

3. Statistical Significance & Data Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

No ML evaluation results found.

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	17.99	10.38	122.8	1001.0	0.1184	0.2776	0.3001	0.1471	0.2419	0.07871
2	20.57	17.77	132.9	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667
3	19.69	21.25	130.0	1203.0	0.1096	0.1599	0.1974	0.1279	0.2069	0.05999
4	11.42	20.38	77.58	386.1	0.1425	0.2839	0.2414	0.1052	0.2597	0.09744
5	20.29	14.34	135.1	1297.0	0.1003	0.1328	0.198	0.1043	0.1809	0.05883
6	12.45	15.7	82.57	477.1	0.1278	0.17	0.1578	0.08089	0.2087	0.07613
7	18.25	19.98	119.6	1040.0	0.09463	0.109	0.1127	0.074	0.1794	0.05742
8	13.71	20.83	90.2	577.9	0.1189	0.1645	0.09366	0.05985	0.2196	0.07451
9	13.0	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389
10	12.46	24.04	83.97	475.9	0.1186	0.2396	0.2273	0.08543	0.203	0.08243
11	16.02	23.24	102.7	797.8	0.08206	0.06669	0.03299	0.03323	0.1528	0.05697
12	15.78	17.89	103.6	781.0	0.0971	0.1292	0.09954	0.06606	0.1842	0.06082
13	19.17	24.8	132.4	1123.0	0.0974	0.2458	0.2065	0.1118	0.2397	0.078
14	15.85	23.95	103.7	782.7	0.08401	0.1002	0.09938	0.05364	0.1847	0.05338
15	13.73	22.61	93.6	578.3	0.1131	0.2293	0.2128	0.08025	0.2069	0.07682
16	14.54	27.54	96.73	658.8	0.1139	0.1595	0.1639	0.07364	0.2303	0.07077
17	14.68	20.13	94.74	684.5	0.09867	0.072	0.07395	0.05259	0.1586	0.05922
18	16.13	20.68	108.1	798.8	0.117	0.2022	0.1722	0.1028	0.2164	0.07356
19	19.81	22.15	130.0	1260.0	0.09831	0.1027	0.1479	0.09498	0.1582	0.05395
20	13.54	14.36	87.46	566.3	0.09779	0.08129	0.06664	0.04781	0.1885	0.05766